PV101 Day 8

Cumulative review, 25-question final, reflection, and closeout

SHOW WHAT YOU KNOW.

Safety first. Units count. Evidence beats labels. Work independently during the graded final.

PV101 is complete at the end of today.

Today's flow

Readiness check -> cumulative review -> final -> reflection -> closeout



The morning quiz is a short readiness check, not the final.
The final runs 1:00-2:10 and is scored out of 25.
Reflection, evidence, and cleanup formally close PV101.

The book progression

Two chapters per assigned reading day

Day 1	Day 2	Day 3	Day 4	Day 5	Day 6
Ch. 1-2	Ch. 3-4	Ch. 5-6	Ch. 7-8	Ch. 9-10	Ch. 11-12

Final: exactly two questions from each chapter, then one cumulative troubleshooting scenario.

Review map

Six connected skill groups

Energy path
source -• load

Electricity
V, A, W, Wh

Measurement
meter ✦ expected value

Workmanship
mechanical ✦ labels

Protection
fuses ✦ disconnects

Sizing
loads -• system

Core formulas

Write the formula, substitute values, keep the unit

Power
W = V x A

Energy
Wh = W x h

Battery capacity
Ah = Wh / V

Array estimate
W = Wh / PSH

1 kW = 1,000 W.
Series voltage adds; matched parallel current adds.
A number without its unit is incomplete evidence.

Meter discipline

The meter answers a specific question

1	**Question**	What do I expect?
2	**Function**	VDC, VAC, ohms, continuity?
3	**Ports ✦ range**	Leads correct?
4	**Energy state**	Live voltage or de-energized continuity?
5	**Probe points**	Exactly where?
6	**Evidence**	Expected vs actual

Workmanship is electrical quality

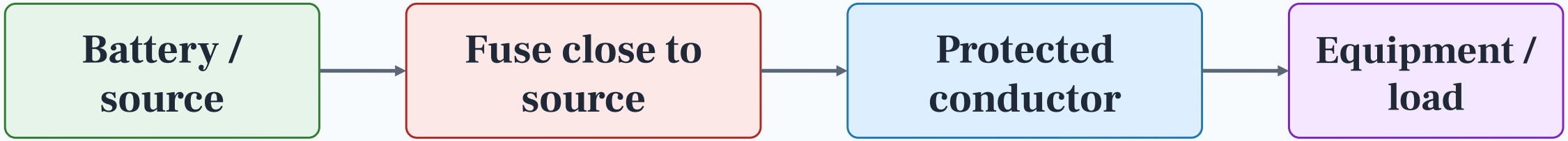
A good circuit must also be serviceable

Correct stripping and crimping; no nicked strands or exposed copper.
Tug-test terminations and support conductors mechanically.
Route for service without strain or confusion.
Label conductors, protection, disconnects, and test points.
Document the restored known-good state after work.

Cleanup and documentation are part of the technical task.

Protection: follow the source

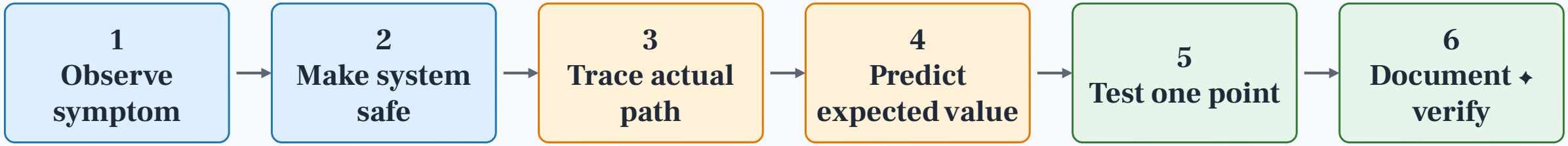
A fuse protects the conductor; a disconnect creates a controllable state



Never bypass protection to “see if it works.”
A label is a clue, not proof.
After replacement, find the cause and verify restoration.

Troubleshooting chain

One test should separate likely causes



Do not replace several parts at once. That destroys the evidence trail.

Series and parallel connections

Keep voltage, current, and documented equipment limits separate

Series
voltage adds
current stays limited

Parallel
current adds
voltage stays matched

Check ratings
from actual labels
and instructions

Connection rules support diagnosis; equipment boundaries still come from documented ratings.

Sizing review

Loads first, assumptions visible

1	**List W ✦ hours**	Show formula + unit
2	**Calculate Wh/day**	Show formula + unit
3	**Apply autonomy ✦ usable fraction**	Show formula + unit
4	**Convert Wh to Ah**	Show formula + unit
5	**Use PSH ✦ loss input**	Show formula + unit
6	**Check concurrent W ✦ surge**	Select inverter from power

Final assessment

25 questions | 25 points | traditional graded test

16 multiple choice

4 calculations

2 diagram / sequence

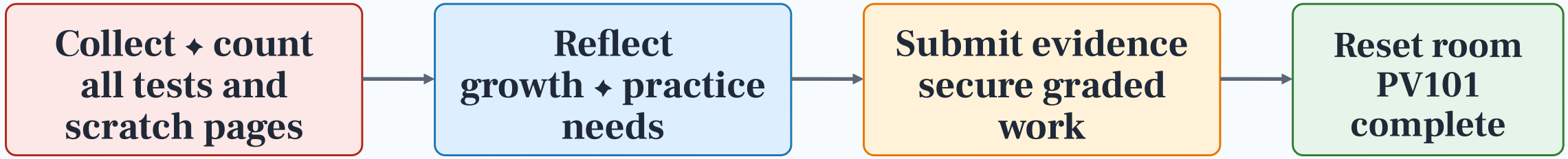
2 short explanations

1 integrative scenario

Work independently.
Show formulas and units.
Ask for clarification of directions only.
Turn in the final before starting reflection.

After the final

Collection -> reflection -> evidence -> closeout



No mandatory correction or retest cycle. Note issues for instructor review.

PV101 complete

Close the course simply and professionally

YOU COMPLETED PV101.

Safety. Measurement. Workmanship. Protection. Troubleshooting. Sizing.

Reflection submitted. Materials counted. Tests secured. Tables clear.